ANCILLARY REVIEWS

DO NOT DELETE. Submit the completed checklist below with your protocol.

Which ancillary reviews do I need and when do I need them? Refer to HRP-309 for more information about these ancillary reviews.			
Select yes or no	**Does your study…**	*If yes…*	***Impact on IRB Review***
☐ **Yes** ☒ **No**	Include Gillette Children's Specialty Healthcare resources, staff or locations?	*Gillette Scientific Review and Gillette Research Administration approval is required. Contact: research@gillettechildrens.com*	**Approval must be received prior to IRB committee/ designated review.** **Consider seeking approval prior to IRB submission.**
☐ **Yes** ☒ **No**	Involve Epic, or Fairview patients, staff, locations, or resources?	*The Fairview ancillary review will be assigned by IRB staff. Contact: ancillaryreview@Fairview.org*	
☐ **Yes** ☒ **No**	Include evaluation of drugs, devices, biologics, tobacco, or dietary supplements or data subject to FDA inspection?	*STOP – Complete the Medical Template Protocol (HRP-590).*	
☐ **Yes** ☒ **No**	Require Scientific Review? Not sure? See guidance in the Investigator Manual (HRP-103).	Scientific review will be conducted by the panel that reviews your study.	
☐ **Yes** ☒ **No**	Relate to cancer patients, cancer treatments, cancer screening/prevention, or tobacco	*Complete the CPRC application process.* *Contact: ccprc@umn.edu*	
☐ **Yes** ☒ **No**	Have a PI or study team member with a conflict of interest?	*The COI ancillary review will be assigned by IRB staff.* *Contact: becca002@umn.edu*	
☐ **Yes** ☒ **No**	Include the use of radiation? (x-ray imaging, radiopharmaceuticals, external beam or brachytherapy)	*STOP – Complete the Medical Template Protocol (HRP-590).*	**Approval from these committees must be received prior to IRB approval;** **These groups each**
☐ **Yes** ☒ **No**	Use the Center for Magnetic Resonance Research (CMRR) as a study location or MR at Masonic Institute for the Developing Brain (MIDB) as a study location?	*Complete the CMRR pre-IRB ancillary review* *Contact: ande2445@umn.edu*	

☐ **Yes** ☒ **No**	Include the use of recombinant or synthetic nucleic acids, toxins, or infectious agents?	*STOP – Complete the Medical Template Protocol (HRP-590).*	**have their own application process.**
☐ **Yes** ☒ **No**	Include the use of human fetal tissue, human embryos, or embryonic stem cells?	*STOP – Complete the Medical Template Protocol (HRP-590)*	
☐ **Yes** ☒ **No**	Include use of protected health information (PHI)? OR Include international collaborators and involves the collection, transmission, and storage of health data?	*If yes, HIPCO may conduct a review of this protocol. This review will be assigned by IRB staff.* *Contact: privacy@umn.edu*	**Approval must be received prior to IRB approval.** **These groups do not have a separate application process but additional information from the study team may be required.**
☐ **Yes** ☒ **No**	Include the use of a controlled substance?	*STOP – Complete the Medical Template Protocol (HRP-590).*	
☐ **Yes** ☒ **No**	Use data from CTSI Best Practices Integrated Informatics Core (BPIC), formerly the AHC Information Exchange (IE)?	*The CTSI Best Practices Integrated Informatics Core ancillary review will be assigned by IRB staff.* *Contact: bpic@umn.edu*	
☐ **Yes** ☒ **No**	Plan to use CTSI Monitoring services, and/or have an IND, IDE, or designated NSR-IDE by the UMN IRB?	*The CTSI monitoring ancillary review will be assigned by IRB staff.* *Please note eligibility criteria here.* *Contact: fencl003@umn.edu*	
☐ **Yes** ☒ **No**	Use histology or procurement support from Biospecimen Support, formerly Biorepository and Laboratory Services (BLS) or Bionet)?	*STOP – Complete the Medical Template Protocol (HRP-590)*	
☐ **Yes** ☒ **No**	Need to be registered on clinicaltrials.gov?	*If you select "No" in ETHOS, the clinicaltrials.gov ancillary review will be assigned by IRB staff.* *Contact:* fencl003@umn.edu	

☐ **Yes** ☒ **No**	Use of student education records?	*Assigned if FERPA guidance is needed during IRB review.* *Contact ande8670@umn.edu*	
☐ **Yes** ☒ **No**	Does your research include collaborations with Tribal partners, Tribal communities, Tribal-serving institutions, or include focused recruitment of Indigenous Peoples?	*The Office of Native American Affairs ancillary review may be assigned.* *See University of Minnesota Guidelines for Indigenous Research.*	**May not impact IRB review/approval.**
☐ **Yes** ☒ **No**	Require registration in OnCore?	*If you select "No" or "I Don't Know" in ETHOS, the OnCore ancillary review will be assigned by IRB staff.* *Contact: oncore@umn.edu*	**Does not affect IRB approval.**
☐ **Yes** ☒ **No**	Propose use of eConsent process that captures electronic signatures of participants?	*REDCap Ancillary Review will be assigned to confirm IRB approval status prior to moving your eConsent to production in REDCap.* *Contact redcap@umn.edu*	
☐ **Yes** ☒ **No**	Propose to use Community-University Health Care Center (CUHCC) resources or include access to patients or their data?	*Contact hlogren@uumn.edu*	

PROTOCOL COVER PAGE

Protocol Title	Multimodal Anthropomorphism in Companion Chatbots: Examining Avatar Choice and Adaptive Language Style with Kagami
Principal Investigator/Faculty Advisor	Name: Dr. Cecilia Xi Wang
	Affiliation: ☒ UMN ☐ Fairview ☐ Gillette
	UMN Home Department: DHA UX & Graphic Design
	UMN Home Dept ID: 10830
	Telephone Number: 612-624-9700
	Email Address: ceciw@umn.edu
Student Investigator	Name: James Brandt
	Current Academic Status (Student, Fellow, Resident): Student
	Department: Design Innovation - 12186
	Telephone Number: 206-636-4098
	Institutional Email Address: bran1400@umn.edu
Scientific Assessment	I believe Scientific Assessment is not required.
Version Number/Date:	1.1 – 6/26/25

REVISION HISTORY

Describe changes made to the protocol after initial approval.

Revision #	Version Date	Summary of Changes	Consent Change?

NOTE: Leave this section blank for the initial submission. The revision history should be documented for modifications to approved studies.

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ABBREVIATIONS/DEFINITIONS

AI: Artificial Intelligence. Refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions.

AIGC: AI-Generated Content. Refers to content, such as images or text, created by artificial intelligence systems.

ANOVA: Analysis of Variance. A statistical test used to compare the means of two or more groups.

API: Application Programming Interface. A set of rules and protocols that allows different software applications to communicate with each other.

Chatbot: A computer program designed to simulate conversation with human users, especially over the Internet.

DVs: Dependent Variables. In experimental research, these are the variables being measured or tested, which are expected to change as a result of manipulations of the independent variables. In this study, DVs include perceived personalization, trust, rapport, social presence, engagement, and satisfaction.

eIC: Electronic Informed Consent. The process of obtaining informed consent using electronic systems and media.

FastAPI: A modern, fast (high-performance) web framework for building APIs with Python.

GPT: Generative Pre-trained Transformer. A type of large language model developed by OpenAI (e.g., GPT-4.1-nano, GPT-Image-1).

HAI: Human-AI Interaction. A subfield of Human-Computer Interaction that focuses on the study of interactions between humans and artificial intelligence systems.

HCI: Human-Computer Interaction. A multidisciplinary field of study focusing on the design of computer technology and, in particular, the interaction between humans (the users) and computers.

HRPP: Human Research Protections Program. The office at an institution responsible for overseeing the ethical conduct of research involving human participants.

HST: Health Science Technology. Refers to IT infrastructure and devices managed by UMN's Health Sciences for handling sensitive data, often including PHI.

IRB: Institutional Review Board. A committee that performs ethical review of proposed research involving human subjects to ensure the protection of their rights and welfare.

IVs: Independent Variables. In experimental research, these are the variables that are manipulated or changed by the researcher to observe their effect on the dependent variables. In this study, IVs are Avatar Type and LSM Implementation.

LIWC: Linguistic Inquiry and Word Count. A text analysis software program that counts words in psychologically meaningful categories.

LLM: Large Language Model. An AI model trained on vast amounts of text data to understand, generate, and manipulate human language.

LSM: Language Style Matching. The nonconscious tendency for individuals to match or converge on each other's linguistic styles (e.g., use of function words) during conversation.

NLTK: Natural Language Toolkit. A suite of libraries and programs for symbolic and statistical natural language processing for English written in the Python programming language.

OpenAI: An AI research and deployment company, provider of the GPT models used in this study.

PI: Principal Investigator. The lead researcher responsible for the design, conduct, and oversight of a research study.

Prolific: An online platform used to recruit participants for research studies.

React: A JavaScript library for building user interfaces.

SOP: Standard Operating Procedure. A set of step-by-step instructions compiled by an organization to help workers carry out complex routine operations.

UMN: University of Minnesota.

1.0 Objectives

1.1 Purpose:

This study aims to empirically investigate the independent and interactive effects of two key multimodal anthropomorphic design features in companion-style AI chatbots: (1) user-controlled avatar representation (specifically, the choice between selecting a premade avatar, generating a unique avatar via AI image generation, or having no avatar) and (2) dynamic, adaptive language style matching (LSM) (adaptive style vs. static style). The primary purpose is to understand how these features influence users' relational perceptions of the chatbot, including perceived personalization, trust, rapport, social presence, engagement, and overall satisfaction during an open-ended, non-task-based conversational interaction with the "Kagami" chatbot.

2.0 Background:

2.1 Significance of Research Question/Purpose:

The deployment of conversational AI systems, particularly as AI companions designed for social and emotional support, is rapidly increasing (Skjuve et al., 2023; Chandra et al., 2025). While anthropomorphism is recognized as a key factor in shaping user perceptions (Van Pinxteren et al., 2020; Blut et al., 2021), there is limited research on how combined multimodal cues (visual and linguistic) and user agency in personalization affect the formation of human-chatbot relationships in non-task-based contexts. Specifically, research on dynamic Language Style Matching (LSM) in LLM-powered agents (Postmus, 2023) and the comparative value of user-initiated avatar customization (e.g., AI-generated avatars) versus system-provided options or no visual representation is scarce (Schlimbach et al., 2023; Tan, 2024). This study addresses this gap by examining the interplay of user-driven visual anthropomorphism and adaptive linguistic style, offering insights crucial for the responsible design and deployment of engaging and ethically-sound relational AI tools that can genuinely contribute to human well-being (De Freitas et al., 2024).

2.2 Preliminary Data:

A preliminary benchmarking study was conducted on ten Large Language Models (LLMs) to assess their inherent language style matching (LSM) capabilities. An ANOVA revealed statistically significant differences in LSM performance among these models ($p = .0467$). Based on an optimal balance of LSM performance, low latency, and cost-efficiency, OpenAI's GPT-4.1-nano was selected as the core dialogue engine for the "Kagami" chatbot in this experiment

2.3 Existing Literature:

This research draws upon theories of anthropomorphism in HCI (Van Pinxteren et al., 2020), which explains how human-like cues make agents seem more familiar and relatable (Blut et al., 2021). It incorporates findings on visual anthropomorphism via avatars, including the Proteus Effect (Yee & Bailenson, 2007) where avatar

characteristics influence user behavior and enhance social presence (Jin, 2011; Peña et al., 2024), and the emerging potential of AI-generated content (AIGC) avatars (Tan, 2024). Furthermore, it builds on research into linguistic anthropomorphism, particularly Language Style Matching (LSM). LSM, the nonconscious convergence of linguistic patterns, is linked to rapport, trust, and relationship stability in human-human interaction (Ireland & Pennebaker, 2010; Ireland et al., 2011; Colton et al., 2024). While static linguistic styles in chatbots have been studied (Liebrecht et al., 2020), research on implementing and evaluating dynamic, adaptive LSM in LLM-based agents remains under-researched (Postmus, 2023; Wang, Theune, & Srivastava, 2023). The study also considers literature on user agency and personalization in value co-creation with AI (Schlimbach et al., 2023). Systematic literature reviews highlight a focus on text-based chatbots and task efficiency, often overlooking multimodal features and relational dynamics in non-task-based AI (Rapp et al., 2021).

3.0 Research questions/hypothesis

3.1 Primary Research Question/Hypothesis:

1. Research Questions
    a. How do different types of user-chosen visual anthropomorphism (selecting a premade avatar, generating a unique avatar via OpenAI GPT-Image-1, or no avatar) influence user perceptions (perceived personalization, trust, rapport, social presence, engagement, satisfaction) toward the "Kagami" companion chatbot?
    b. How does dynamic, LLM-tuned Language Style Matching (LSM) (adaptive LSM adapting to user style vs. static LSM using a consistent style) influence these same user perceptions and experience outcomes?
    c. How do user-chosen visual anthropomorphism and dynamic LSM interact to influence these user perceptions and overall user experience outcomes during open-ended conversation?
2. Hypotheses:
    a. H1: The presence of an avatar (either "Select Premade" or "Generate Unique Avatar") will result in significantly higher scores on perceived personalization, trust, rapport, social presence, engagement, and satisfaction compared to the "No Avatar" condition.
    b. H2: The "Adaptive LSM" condition will result in significantly higher scores on perceived personalization, trust, rapport, social presence, engagement, and satisfaction compared to the "Static LSM" condition.
    c. H3 (Congruence Hypothesis): We hypothesize a significant interaction effect between Avatar Type and LSM Implementation. This is based on a congruence hypothesis, where the alignment between high user agency in the visual modality ('Generate Unique Avatar') and high system responsiveness in the linguistic modality ('Adaptive LSM') is expected to

create a synergistic effect. Following research showing that the combination of congruent visual and conversational cues enhances the perceived humanness of chatbots (Go & Sundar, 2019), we predict this condition will yield the highest scores across all dependent variables.

3.2 Secondary Research Question(s)/Hypothesis:

No distinct secondary research questions or hypotheses are proposed at this time. Exploratory analyses may investigate the mediating roles of social presence and perceived personalization on downstream outcomes, and examine relationships between behavioral log data (e.g., message counts) and self-reported measures.

4.0 Study Intervention(s)/Interaction(s):

4.1 Description:

This interaction is framed as a casual, friendly chat. To ensure a consistent baseline for all interactions, the conversation is initiated with an identical, hardcoded greeting for all participants: "Hey there, I'm Kagami. What's on your mind today, or how's your day been so far?" The system does not proactively suggest conversational topics but employs mechanisms to maintain conversational flow, which differ by experimental condition as described below.

The experimental manipulations are:

1. **Avatar Type (Independent Variable 1):**
    a. **Select Premade Avatar**: Participants choose an avatar from a set of seven predefined options. This set was designed according to several core principles to ensure controlled variation while minimizing confounding variables:
        i. **Thematic & Stylistic Cohesion:** All avatars are non-human, animal-themed characters rendered in a consistent art style to avoid biases associated with human representations.
        ii. **Controlled Variation:** The avatars vary by common animal species but maintain a similar neutral-to-positive facial expression and posture.
        iii. **Agency:** The set provides a sufficient variety of options to ensure the participant feels a sense of meaningful choice.
    b. **Generate Unique Avatar:** Participants provide text prompts to OpenAI's GPT-Image-1 API to generate a unique avatar for the chatbot.
    c. **No Avatar**: Participants interact with the chatbot without any visual avatar representation.
2. **LSM Implementation (Independent Variable 2):**
    a. **Adaptive LSM:** The Kagami chatbot, powered by OpenAI's GPT-4.1-nano, will dynamically adapt its linguistic style to match the user's style. This adaptation is achieved via a custom Python-based pipeline grounded in

established communication frameworks. For each conversational turn, the pipeline analyzes the user's input along several key linguistic dimensions:

i. **Core LSM (LIWC-based)**: It measures the relative frequency of key function word categories (e.g., pronouns, articles, negations), based on the foundational framework of Pennebaker's Linguistic Inquiry and Word Count (LIWC).
ii. **Broader Stylistic Traits**: It analyzes approximately 30 additional style markers, including formality, sentiment, and linguistic complexity, using a combination of transformer models and established text-analysis libraries.
iii. **Interactional Patterns**: It tracks conversational behaviors, such as the user's inquisitiveness (i.e., frequency of question-asking).
iv. These dimensions are then used to dynamically construct a system prompt for the LLM, instructing it to generate a response that is stylistically congruent with the user's own language use.

b. **Static LSM:** The Kagami chatbot will use a consistent, predefined linguistic style, without adapting to the user's specific style.

Participants are randomly assigned to one of six conditions, each representing a unique combination of an Avatar Type level and an LSM Implementation level. The interaction occurs via a custom-built web application (React frontend, Python/FastAPI backend).

Data Privacy and OpenAI Usage:
Information entered into the Kagami chatbot and the avatar creation system will not be accessible to the broader internet community. All interactions are contained within the secure Kagami platform.
Regarding OpenAI's usage: As per OpenAI's API data usage policy (effective March 1, 2023), data sent to the specific API endpoints used in this study (e.g., /v1/chat/completions for GPT-4.1-nano and /v1/images/edits for GPT-Image-1) is not used to train or improve OpenAI's models. OpenAI retains API inputs and outputs for up to 30 days for abuse monitoring purposes only. Critically, no identifiable participant data collected in Qualtrics (e.g., demographics, Prolific ID) will be transmitted directly to OpenAI through these API calls.

5.0 Procedures Involved

5.1 Study Design:

This study will utilize a 3x2 between-subjects factorial experimental design.

1. Factor 1 (Avatar Type): 3 levels (Select Premade Avatar, Generate Unique Avatar, No Avatar)
2. Factor 2 (LSM Implementation): 2 levels (Adaptive LSM, Static LSM)

This design results in six independent experimental conditions. Participants will be randomly assigned to only one condition.

5.2 Study Procedures:

1. Informed Consent: Participants recruited via Prolific will first be presented with an electronic informed consent form on Qualtrics. Agreement is required to proceed.
2. Pre-Survey: Participants will complete a brief pre-survey via Qualtrics, collecting demographics and potentially baseline measures (e.g., tech attitudes).
3. Random Assignment: Participants will be randomly assigned by Qualtrics randomization to one of the six experimental conditions.
4. Website Disclaimer: After random assignment, participants will be redirected to the Kagami Chat website via Qualtrics survey termination ("https://kagami.chat/?PROLIFIC_PID=[prolific_id]&avatar=[avatar_type]&lsm=[lsm_type]&responseid=[qualtrics_response_id]") where they will be provided with another disclaimer.
5. Avatar Interaction (Conditional):
    a. Select Premade Avatar: Participants choose an avatar from predefined options. Data Collected: The ID of the selected premade avatar.
    b. Generate Unique Avatar: Participants use text prompts to generate an avatar via GPT-Image-1. Data Collected: The user-provided text prompt and the URL of the final generated avatar image.
    c. No Avatar: Participants bypass this step.
6. Chatbot Conversation: All participants engage in a 10-minute open-ended conversation with the Kagami chatbot (OpenAI's GPT-4.1-nano). Data Collected: A complete log of the conversation, including the content of each user and bot message, timestamps for each message, and a turn-by-turn message count.
7. Post-Survey: Participants complete a Qualtrics post-survey measuring dependent variables and manipulation checks.
8. Debriefing: Participants receive a final debriefing explaining the study's purpose.
9. Compensation: Participants are redirected to Prolific for payment.

5.3 Study Duration:

- Individual participation: 15–20 minutes.
- Recruitment and data collection: ~1 week (N = 162 participants).

5.4 Follow-Up:

N/A: There is no planned follow-up with participants after their single interaction session and post-survey completion.

6.0 Sharing of Results with Participants

6.1 Sharing Results:

Participants will not receive individual results or feedback from their interaction. However, a summary of the aggregated, de-identified findings of the overall study may be made available to participants upon request after the completion of the Master's thesis and any subsequent publications. The debriefing statement provided at the end of the study will include contact information for the Principal Investigator,

whom participants can email to request this summary. Any shared summary will focus on general trends and will not contain any information that could identify individual participants.

7.0 Study Population

7.1 Inclusion Criteria:

- Adults aged 18 years or older.
- Fluent in English (self-reported, and as a pre-screening option on Prolific).
- Recruited via the Prolific online research platform.
- Have access to a computer or mobile device with a stable internet connection and a compatible web browser to access the online experiment.

7.2 Exclusion Criteria:

- Are under 18 years of age.
- Are not fluent in English.
- Have previously participated in pilot testing for this specific "Kagami" experiment.
- Are unable to provide informed consent.

7.3 Screening:

Primary screening will occur through Prolific's built-in pre-screening filters, which allow researchers to target participants based on age (18+) and self-reported English language fluency. The informed consent form will also reiterate the age requirement (18+ years). No other active screening procedures will be conducted by the research team beyond these platform-based filters and the consent process.

8.0 Vulnerable Populations

University requirements for inclusion of vulnerable populations may be stricter than what may be acceptable for sponsors or for lead investigators at other institutions.

8.1 Vulnerable Populations:

Population / Group	Identify whether any of the following populations will be focus of the research (targeted), included, but not necessarily the focus or excluded from participation in the study.
Children	Excluded
Pregnant women/persons	included but not the focus
Fetuses	Excluded

Neonates	Excluded
Prisoners	Excluded.
Adults lacking capacity to consent and/or adults with diminished capacity to consent	Excluded
Non-English speakers	Excluded
Those unable to read (illiterate)	Excluded
Employees of the researcher	included but not the focus
Students of the researcher	included but not the focus
Undervalued or disenfranchised social group	included but not the focus.
Active members of the military (service members), DoD personnel (including civilian employees)	included but not the focus.
Individual or group that is approached for participation in research during a stressful situation such as emergency room setting, childbirth (labor), etc.	Excluded
Individual or group that is disadvantaged in the distribution of social goods and services such as income, housing, or healthcare.	included but not the focus
Individual or group with a serious health condition for which there are no satisfactory standard treatments.	Excluded
Individual or group with a fear of negative consequences for not participating in the research (e.g. institutionalization, deportation, disclosure of stigmatizing behavior).	Excluded

Any other circumstance/dynamic that could increase vulnerability to coercion or exploitation that might influence consent to research or decision to continue in research.	Excluded.

8.2 Additional Safeguards, if any, to ensure inclusion is appropriate:
Given that the study recruits a general adult population through Prolific and actively excludes most vulnerable populations listed, extensive additional safeguards beyond standard ethical research practices are not deemed necessary. Standard safeguards being employed include:

8.2.1 Informed Consent: A clear and comprehensive informed consent process outlining all aspects of the study, voluntary participation, and the right to withdraw.

8.2.2 Anonymity/Confidentiality: Data will be de-identified to the extent possible, and Prolific IDs will be separated from the research data after compensation.

8.2.3 Voluntary Participation: Participants are free to decline or withdraw at any point without penalty.

8.2.4 Fair Compensation: Participants are compensated for their time and effort at a rate consistent with Prolific guidelines.

8.2.5 Minimal Risk Design: The study procedures involve minimal risk, comparable to everyday online interactions.

8.2.6 For populations marked as "included but not the focus" (e.g., pregnant persons, employees/students of the researcher (if incidentally recruited via Prolific), members of undervalued groups), these standard safeguards are considered sufficient as their inclusion is incidental and not targeted, and the research procedures do not pose specific or increased risks to these groups beyond those for the general adult population.

8.3 Rationale for exclusions in the table above if the research has the potential for direct benefit:
This research is not designed to offer direct benefits to individual participants. The exclusions are primarily based on:

8.3.1 Ethical Considerations and Scope:

8.3.1.1 Children, prisoners, adults lacking capacity to consent: These populations are excluded as per standard ethical guidelines for research that is not specifically designed to study or benefit them, and which would necessitate complex additional safeguards (e.g., parental consent, assent, advocate involvement) beyond the scope and minimal risk nature of this study.

8.3.1.2 Individuals in stressful situations or with serious health conditions: The study is not equipped to handle disclosures or needs that might arise from these populations, and the non-therapeutic nature of the AI companion could be misleading or unhelpful.

8.3.2 Practical and Methodological Considerations:

8.3.2.1 Non-English speakers: The chatbot interaction, survey instruments, and consent materials are in English, making participation by non-English speakers impractical.

8.3.2.2 Those unable to read (illiterate): The online text-based nature of the study (consent, surveys, chat) requires literacy.

8.3.2.3 The research aims to contribute to generalizable knowledge about human-AI interaction, which may indirectly benefit society by informing the design of better AI tools, but direct benefit to excluded individuals from participation in this specific study is not anticipated.is not considered a direct benefit for the purposes of this section.

9.0 Number of Participants at this Site

9.1 Number of Participants to be Consented by Researchers at this Site:

Approximately 162 participants will be recruited and consented for this study. This sample size is based on a power analysis (G*Power, f = .25, power = .80) indicating N ≈ 162 is needed to detect medium effects for the 3×2 ANOVA. The study is conducted online and is considered a single-site study managed by the University of Minnesota.

10.0 Local Recruitment Methods

10.1 Recruitment Process:

Participants will be recruited exclusively through the Prolific online research platform (www.prolific.com). A study listing will be created on Prolific, visible to eligible platform users. Interested individuals who meet the pre-screening criteria (age 18+, English fluency) can select the study to participate. Upon selecting the study, they will be directed to the initial study page containing the informed consent form.

10.2 Recruitment Materials:

The primary recruitment material will be the study listing posted on the Prolific platform. This listing will include:

10.2.1 A concise and non-misleading study title (e.g., "Chat with an AI Companion and Share Your Feedback").

10.2.2 A brief description of the study tasks (e.g., "You will chat with an AI for 10 minutes, answer some questions about your experience, and complete short surveys.").

10.2.3 The estimated duration of participation (15–20 minutes).

10.2.4 The compensation amount.

10.2.5 Any specific eligibility criteria not covered by Prolific's general pre-screeners (though none are planned beyond age/language).

10.2.6 A copy of the proposed Prolific study listing text will be provided to the IRB as an appendix/attachment.

.

10.3 Payment:

Participants will be compensated approximately $3.50 (reflecting an hourly rate of approximately $10.50/hour for an estimated 15–20 minutes of participation) upon successful completion of the study. Payment will be processed through their Prolific account. Compensation is provided for their time and effort and is not contingent on their specific responses or performance in the study.

10.4 Recruitment Barriers and Challenges

Potential recruitment barriers include:

10.4.1 Slower-than-anticipated uptake on Prolific: This could be due to competition from other studies or insufficient participant pool matching the pre-screeners at certain times.

10.4.2 Achieving balanced group sizes: While randomization aims for balance across the six conditions, slight imbalances may occur due to the natural flow of recruitment.

10.4.3 Participant attrition: Some participants might start the study but not complete it (e.g., drop off during the pre-survey or chat).

10.4.4 The online nature of the study, while broadly accessible, may exclude individuals without reliable internet access or digital literacy, which can disproportionately affect certain underserved populations. Recruitment via Prolific aims to reach a broad adult audience.

10.5 Recruitment Monitoring and Mitigations plans

10.5.1 Monitoring:

Recruitment progress (number of participants, completion rates) will be monitored daily via the Prolific dashboard and Qualtrics. A hard quota of 162 completes is enforced in Qualtrics; CAPTCHA and ‘Prevent Ballot Box Stuffing’ are enabled to mitigate automated entries.

10.5.2 Mitigation for Slow Uptake:

If recruitment is slow, the study description on Prolific may be refined for clarity, or the compensation rate may be slightly increased (within ethical limits and after IRB notification/amendment if significant). The study may also be "boosted" on Prolific if such features are available and appropriate.

10.5.3 Mitigation for Imbalanced Groups:

While the randomization should handle this, the study will aim to recruit slightly more than the minimum N per cell if initial recruitment suggests potential for slight imbalances. The final sample size target of 162 allows for some flexibility.

10.5.4 Mitigation for Attrition:

Clear instructions, a user-friendly interface, and the relatively short duration are designed to minimize attrition. Data from incomplete participations will typically not be included in the final analysis, and recruitment will continue until the target number of completed participations is reached.

11.0 Withdrawal of Participants

11.1 Withdrawal Circumstances:

Participants may withdraw voluntarily from the study or the PI may withdraw a Participants may choose to withdraw from the study at any point before, during, or immediately after their participation, without any penalty or loss of benefits to which they would otherwise be entitled (though compensation is typically for completed studies on Prolific, partial compensation may be handled per Prolific policy if significant time was invested). Reasons for withdrawal are not required.

11.2 Withdrawal Procedures:

During the study: Participants may withdraw from the study at any time. Data from incomplete participations (e.g., from participants who close the browser or "return" the study on Prolific) will not be included in the final dataset for analysis. If a participant completes the entire study but later wishes to have their data withdrawn, they may contact the PI. The research team will make all reasonable efforts to remove their data, provided the request is made before the final dataset has been fully de-identified and aggregated for analysis.

After completion (data withdrawal): If a participant completes the study but later wishes to have their data withdrawn, they can contact the Principal Investigator using the contact information provided in the consent and debriefing forms. The research team will make reasonable efforts to remove their data from the dataset, provided the request is made before data has been fully de-identified and aggregated for analysis, and the data can be uniquely identified (e.g., via their Prolific ID or the unique participant code passed in the URL).

Prolific has its own mechanisms for participants returning a study submission; if a participant does this, their data will not be used.

11.3 Termination Procedures:

The investigator reserves the right to terminate a participant's involvement in the study if there is clear evidence of non-serious participation (e.g., providing nonsensical or random text inputs during the chat, attempting to use automated scripts) or if technical issues on the participant's end prevent meaningful data collection. In such rare cases, the participant would typically not be compensated for

a non-valid submission as per Prolific's terms. This will be handled on a case-by-case basis, prioritizing fairness to the participant.

11.3.1 There are no specific pre-planned circumstances for early termination of the entire study. However, the study could be terminated prematurely by the PI in consultation with the IRB if significant unanticipated risks emerge, major technical failures prevent data collection, or if it becomes clear that recruitment goals cannot be met despite mitigation efforts. In such a case, any collected data would be handled as per IRB guidance.

11.4 Identify Potential Barriers to Ongoing Participation – Retention Monitoring and Mitigation Plans

Ongoing participation is limited to a single 15–20 minute session.

1. **Potential Barriers:**
   - **a.** Technical difficulties (e.g., slow internet, browser incompatibility, though the web app will be tested for common browsers).
   - **b.** Boredom or disengagement during the 10-minute chat.
   - **c.** Frustration with the AI's responses.
2. **Retention Monitoring (within session):**
   - **a.** The system will log timestamps, allowing for identification of very short interaction durations.
3. **Mitigation Plans:**
   - **a.** The web application will be designed to be robust and user-friendly.
   - **b.** The chat interaction is designed to be companionable and open-ended to maintain interest.
   - **c.** The 10-minute duration is relatively short to minimize fatigue.
   - **d.** Clear instructions will be provided at each step.

12.0 Risks to Participants

12.1 Foreseeable Risks:

The risks associated with participation in this study are considered minimal, meaning they are no greater than those ordinarily encountered in daily life when using the internet or engaging in online conversations.

1. **Minor psychological discomfort or boredom:** Participants might experience slight boredom during the chat or surveys, or mild frustration if the chatbot's responses are not ideal or if they encounter technical glitches. The open-ended nature of the chat could also lead participants to briefly touch upon mildly personal topics, though the AI is not designed to delve deeply into sensitive issues and, using prompt engineering, steers users away from sensitive topics ("If the user brings up unrelated topics (like personal advice, medical, legal, or financial matters),

gently steer the conversation back to everyday things like music, shows, or memories.")

2. **Misattribution of AI capabilities/Potential for misplaced trust or emotional connection:** As the study examines anthropomorphism, there's a small risk participants might perceive the AI as more human-like or emotionally capable than it is, potentially leading to brief feelings of connection or slight disappointment when the interaction ends or its AI nature is re-emphasized.
3. **Breach of Confidentiality:** While unlikely given the security measures, there is always a minimal risk of a data breach with any online data collection. This includes the data processed by the research team and data temporarily processed by third-party AI services (OpenAI).
4. **Potential for AI Bias:** The study acknowledges that Large Language Models (LLMs) like GPT-4.1-nano, which powers the Kagami chatbot, may reflect biases present in their extensive training data. While this study does not directly measure or mitigate AI bias, the Kagami chatbot is designed for general, companion-style conversation and is explicitly instructed via prompt engineering to avoid giving advice or engaging in sensitive topics where bias could be more problematic. The focus of this research is on user perception of manipulated visual and linguistic styles, not an evaluation of the AI for specific biases in its informational content.

12.2 Reproduction Risks:

N/A: This study does not involve any interventions, substances, or procedures that would pose risks related to reproduction for any participants.

12.3 Risks to Others:

N/A: No risks to others (i.e., individuals not directly participating in the research) are anticipated as a result of this study. The research involves individual, private online interactions with an AI chatbot and completion of surveys.

13.0 Incomplete Disclosure or Deception

13.1 Incomplete Disclosure or Deception:

There is no deception involved in this study. Participants will be explicitly informed before and during the interaction that they are chatting with an AI ("Kagami").

There will be incomplete disclosure regarding the specific experimental manipulations (i.e., that they are in a condition testing a specific avatar type or a specific language style matching mode) and the detailed research hypotheses until the debriefing stage. This is a standard methodological practice in experimental research to prevent participants from altering their natural behavior or responses in a way that could bias the study results (i.e., to avoid demand characteristics). The debriefing statement provided at the end of the study will fully explain the different conditions and the overall purpose of the research.

14.0 Potential Benefits to Participants

There are no direct benefits anticipated for the individual participants in this study. Participants will receive monetary compensation for their time and effort, as described in Section 10.3.

The potential indirect benefits are to society and the scientific community. This research aims to contribute to a more nuanced understanding of human-AI relationship formation. The findings may:

1. Inform the design of more engaging, effective, and ethically-sound AI companions, educational tools, and digital wellness applications.
2. Provide evidence-based guidance for designers on how multimodal personalization (avatars and language style) and user agency influence user perceptions and engagement.
3. Advance AI literacy by clarifying how AI systems can be designed to foster (or hinder) connection and trust.

15.0 Statistical Considerations

15.1 Data Analysis Plan:

The primary data analysis will involve 3x2 between-subjects Analyses of Variance (ANOVAs) to examine the main effects of Avatar Type (Select Premade vs. Generate Unique vs. No Avatar) and LSM Implementation (Adaptive vs. Static), as well as their interaction effect, on each of the primary dependent variables (perceived personalization, trust, rapport, social presence, engagement, satisfaction, perceived anthropomorphism).

1. If significant main effects or interactions are found, post-hoc tests (e.g., Tukey's HSD) or planned contrasts will be conducted to identify specific differences between condition means.
2. Descriptive statistics (means, standard deviations) will be calculated for all measures. The reliability of multi-item scales will be assessed using Cronbach's alpha.
3. Exploratory analyses may include:
    a. Mediation analyses to explore whether social presence and perceived personalization channel the impact of anthropomorphic cues into downstream outcomes.
    b. Analysis of conversation logs using LIWC-22 to objectively assess LSM and potentially correlate linguistic features with self-reported outcomes.
    c. Exploratory regressions using pre-survey variables as covariates if deemed appropriate.

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15.2 Power Analysis:

A priori power analysis was conducted using G*Power (Faul et al., 2007) to determine the required sample size for the 3x2 ANOVA. To detect a medium effect size ($f = 0.25$) with an alpha of .05 and power of .80, a total sample size of approximately $N = 158$ participants is needed. To account for potential exclusions or incomplete data, the target recruitment is set at 162 participants, aiming for approximately 27 participants per each of the six experimental conditions.

15.3 Statistical Analysis:

Statistical analyses will primarily involve descriptive statistics, reliability analyses (Cronbach's alpha), 3x2 between-subjects ANOVAs, and appropriate post-hoc tests or planned contrasts. All analyses will be conducted using standard statistical software (e.g., SPSS, R). Significance will be assessed at $p < .05$.

15.4 Data Integrity:

Several measures will be taken to ensure data integrity:

1. Data Entry: Survey data will be collected directly into Qualtrics, minimizing manual data entry errors. Chat logs are system-generated.
2. Data Cleaning: Exported data will be reviewed for completeness, outliers, and anomalies. Clear rules for handling missing data (e.g., per-scale rules, listwise deletion for primary analyses if minimal) will be established and documented.
3. Manipulation Checks: Survey items will assess whether participants noticed key aspects of their condition (e.g., presence of an avatar, perceived adaptiveness of language) to ensure the manipulations were effective.
4. Variable Computations: Composite scores for multi-item scales will be calculated according to established scoring procedures.
5. Documentation: All data processing steps and analytic decisions will be documented.

16.0 Confidentiality

16.1 Data Security:

Confidentiality will be protected through the following measures:

1. **Anonymization/De-identification:** Participants will be identified in the Kagami system by a randomly generated participant code passed from Qualtrics, and on Prolific by their anonymous Prolific ID. Prolific IDs are used solely for compensation and initial linking if necessary and will be removed from the final research dataset. No direct personal identifiers (e.g., names, email addresses other than for PI contact) will be collected as part of the research data.
2. **Secure Data Storage:**
    a. Survey data from Qualtrics will be stored on secure University of Minnesota servers.

   b. Chat logs and other data generated by the Kagami application (including avatar choice/prompts, message counts, timestamps) will be stored on secure, password-protected University of Minnesota approved cloud storage (e.g., UMN Enterprise Google Drive with restricted access).

3. **Restricted Access:** Access to identifiable data (during the collection phase) and de-identified research data will be strictly limited to the Student Investigator (James Brandt) and the Faculty Advisor.
4. **Data Transmission:**
   a. Data within UMN systems will be transmitted securely (e.g., HTTPS for web interactions, secure file transfer for data exports).
   b. Chat inputs and avatar generation prompts sent to OpenAI APIs will be transmitted over secure HTTPS connections. As per OpenAI's API data usage policy (effective March 1, 2023), data sent to these API endpoints (/v1/chat/completions with gpt-4.1-nano and /v1/images/edits with gpt-image-1) is not used to train or improve OpenAI's general models (this study will not opt-in to data sharing for training). OpenAI retains API inputs and outputs for up to 30 days for abuse monitoring purposes. No identifiable participant data from Prolific or Qualtrics is sent to OpenAI.
5. **Chat Content and AI Output Review:** While participants are free to discuss various topics, they will be implicitly discouraged from sharing highly sensitive PII through the AI's framing and the study's non-therapeutic nature. Any inadvertently shared PII in chat logs will be removed or masked during data processing if identified, prior to analysis or sharing of de-identified datasets. Similarly, AI-generated text (chatbot responses) will be part of the recorded chat logs and reviewed; while not expected, any inadvertently generated PII or highly unique, potentially re-identifiable conversational patterns attributable to the AI would be addressed during de-identification procedures before wider data sharing.

17.0 Provisions to Monitor the Study to Ensure the Safety of Participants

17.1 Safety Plan:

Given that this study is classified as minimal risk, an extensive safety plan or Data Safety Monitoring Board (DSMB) is not required. Safety will be ensured through:

1. **Ethical Design:** Adherence to the approved IRB protocol, including informed consent and clear risk disclosure.
2. **PI Oversight:** The Student Investigator, under the guidance of the Faculty Advisor, will oversee the study conduct.
3. **AI Safeguards:** The Kagami chatbot includes prompt engineering to prevent generation of harmful content and to gently redirect overly sensitive conversations. The website disclaimer also reinforces the AI's nature.
4. **Voluntary Participation:** Participants can withdraw at any time if they feel uncomfortable.

5. **Contact Information:** PI contact information is provided for any participant concerns.
6. **Unanticipated Problems:** Any unanticipated problems involving risks to subjects or others will be promptly reported to the IRB.

17.2 Data Integrity Monitoring:

Data integrity will be monitored through:

1. Regular checks of the data collection systems (Qualtrics, Kagami backend logs) for proper functioning and data capture.
2. Review of initial datasets for completeness, consistency, and adherence to expected formats.
3. Verification of the randomization process to ensure proper assignment to conditions.
4. Assessment of manipulation check data to confirm experimental manipulations are being perceived as intended.

17.3 Data Safety Monitoring:

For this minimal risk study, formal data safety monitoring by an independent committee is not warranted. The Student Investigator and Faculty Advisor will be responsible for:

1. Periodically reviewing the overall progress of the study.
2. Assessing any participant complaints or issues reported (though none are anticipated beyond minor technical queries).
3. Ensuring that confidentiality and data security measures are maintained throughout the study.
4. The primary focus is on ensuring ethical conduct and data integrity rather than monitoring for adverse safety events typical of clinical trials.

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18.0 Provisions to Protect the Privacy Interests of Participants

18.1 Protecting Privacy:

Privacy will be protected by:

1. **Anonymity in Reporting:** All results will be reported in an aggregated and de-identified format. No individual participants will be identifiable in any reports, presentations, or publications.
2. **Minimization of Identifiable Data:** Only Prolific IDs (for payment) and the system-generated pid are used as temporary identifiers. No names, contact details (other than for PI communication), or other direct PII are collected within the research dataset.
3. **Secure Data Handling:** As described in Section 16.1, data is stored securely with restricted access.
4. **Clear Communication:** The consent form and website disclaimer will inform participants about data collection, use, and confidentiality measures.

5. **No Audio/Video Recording:** The interaction is text-based only..

18.2 Access to Participants:

Participants will not be re-contacted by the research team after the completion of their study session for any reason other than if they initiate contact (e.g., with a question, or to request a summary of findings as described in the debriefing statement). There is no planned follow-up or further data collection from the same participants.

19.0 Compensation for Research-Related Injury

19.1 Compensation for Research-Related Injury:

N/A

19.2 Contract Language:

N/A

20.0 Storing Data for Future Use

20.1 Storage and Access:

De-identified research data (including survey responses, chat logs stripped of any potential identifiers, and derived linguistic metrics) will be stored securely on University of Minnesota approved platforms (e.g., UMN Enterprise Google Drive, secure research servers) for a minimum of three years after the completion of the study or any resulting publications, in accordance with University policy and journal data retention requirements. Access to this stored data will be restricted to the Student Investigator (James Brandt) and the Faculty Advisor.

The de-identified data may be used for future research purposes, such as:

1. Secondary data analyses exploring new research questions related to the current dataset.
2. Meta-analyses combining these findings with other studies.
3. Developing or refining theories of human-AI interaction.
4. Educational purposes (e.g., in future teaching or presentations, always in de-identified and aggregated form).
5. Any new research projects using this data that go beyond the scope of the current protocol will be subject to further IRB review or determination of exemption as appropriate.

Data:

The data to be stored for future use will include:

1. De-identified pre-survey data (demographics, tech attitudes).
2. De-identified post-survey data (responses to scales measuring perceived personalization, trust, rapport, social presence, engagement, satisfaction, perceived anthropomorphism, and manipulation checks).
3. De-identified conversation logs (user and bot utterances, timestamps, message counts).

4. Avatar choice data (selected premade avatar ID, or text prompt used for avatar generation, and potentially the generated image data if stored in a de-identified manner).
5. Linguistic features extracted from chat logs (e.g., LIWC categories, style trait scores).
6. The system-generated unique participant code (pid) to link these data sources, but with Prolific IDs removed.

20.2 Release/Sharing:

The researchers intend to share the findings of this study through scholarly publications (e.g., Master's thesis, peer-reviewed journal articles, conference presentations).

In line with open science practices, a de-identified version of the dataset (excluding any free-text responses that could inadvertently contain PII even after initial scrubbing, or further anonymizing such text) may be made available to other researchers for verification, replication, or secondary analysis. This would typically occur via a public research data repository (e.g., OSF, Zenodo, a university repository) after publication of the primary findings. The consent form will inform participants about this potential for sharing de-identified data for future research purposes. Any data sharing will adhere to University of Minnesota policies and ethical guidelines.

21.0 Consent Process

21.1 Consent Process (when consent will be obtained):

The consent process will occur entirely online at the beginning of the study interaction.

Where: The electronic informed consent form will be presented to participants on the University of Minnesota's Qualtrics platform immediately after they click the study link from Prolific and before any pre-survey questions or study procedures begin.

Waiting Period: Participants can take as much time as needed to read and consider the information presented in the online consent form. There is no mandated waiting period, but the self-paced nature of the online form allows for unhurried review before deciding to proceed.

Determining Understanding: Understanding is inferred by a participant's voluntary affirmative action to consent. The consent form is written in clear, simple language. Participants must actively click a button or check a box labeled "Yes, I consent" (or similar wording) to indicate they have read, understood, and agree to participate. If they select "No, I do not consent," they will be exited from the study, and no data will be collected. Participants are provided with contact information for the study team should they have questions before consenting.

Ongoing Consent: Consent is considered an ongoing process. Participants are reminded of the voluntary nature of their participation in the website disclaimer on Kagami.Chat and in the main consent form. They can withdraw at any time by closing the browser. The debriefing statement will also provide contact information for any post-study questions or concerns.

Documentation & Platform: Written documentation of consent will be obtained electronically using the University of Minnesota's Qualtrics platform, which can capture and timestamp the electronic indication of consent. The UMN HRPP electronic consent template (adapted from HRP-582 Social-Behavioral template) will be used. A copy of the consent form will be made available for participants to download or print for their records.

HIPAA: HIPAA does not apply to this study, so a combined HIPAA Authorization/Consent document is not used.

21.2 Waiver or Alteration of Consent Process (when consent will not be obtained):

N/A

21.3 Waiver of Written/Signed Documentation of Consent (when written/signed consent will not be obtained):

We request a waiver of signed documentation of consent under HRP-411 § 1. The study is conducted entirely online, poses no more than minimal risk, and involves no procedures for which a signature is normally required outside the research context. The protocol does not involve newborn dried blood spots. Participants receive an electronic information sheet (the full consent form) and provide consent by selecting “Yes, I consent” in Qualtrics. Written information describing the research is therefore provided, but no physical or electronic signature is collected.

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21.4 Non-English Speaking Participants:

Only English-speaking participants will be enrolled in this study. All recruitment materials on Prolific will target English speakers, and all study materials (consent form, surveys, chatbot interaction, debriefing) will be in English. No provisions for enrolling non-English speaking participants or for using interpreters are planned.

21.5 Participants Who Are Not Yet Adults (infants, children, teenagers under 18 years of age):

Individuals under the age of 18 years will be excluded from this study. Eligibility criteria on the Prolific recruitment platform will be set to include only adults aged 18

and older. The consent form will also explicitly state that participants must be 18 years or older. No procedures for parental permission or child assent will be used.

21.6 Cognitively Impaired Adults, or adults with fluctuating or diminished capacity to consent:

This study will not enroll adults who are unable to provide consent for themselves. Therefore, procedures for obtaining permission from Legally Authorized Representatives (LARs) or for assessing assent/dissent from the adult unable to consent are not applicable.

21.7 Adults Unable to Consent:

21.7.1 Permission:
N/A

21.7.2 Assent:

N/A

21.7.3 Dissent:
N/A

22.0 Setting

22.1 Research Sites:

Research Activities Location: All research activities will be conducted entirely online.

Participant Identification and Recruitment: Potential participants residing in the United States will be identified and recruited through the Prolific online research platform (www.prolific.com).

Performance of Research Procedures: Participants will complete all study procedures (informed consent, pre-surveys via REDCap, avatar interaction [if applicable], 10-minute chatbot interaction via the Kagami.Chat website, and post-surveys via REDCap) using their own personal computers or mobile devices from locations of their choosing within the United States that have internet access.

Community Advisory Board, School Staff, etc.: No community advisory boards, school boards, or similar external entities are involved in this study's design or conduct.

Research Conducted Outside the Organization: While participants complete the study remotely, the research is designed, managed, and overseen by the research team at the University of Minnesota, with all data stored on UMN-approved platforms..

22.2 International Research:

☐ General Data Protection Regulation (GDPR) applies to this study. Explain: N/A

☐ This research will take place in one or more international locations. Explain: N/A

☐ This research will involve collaborators from outside the United States. Explain: N/A

☐ This research will involve data collection, sharing, access, or transmission between U.S. and international collaborators/institutions. Explain: N/A

22.3 Community Participatory Research:

N/A: This study does not employ a community-based participatory research design and does not partner with specific defined communities.

23.0 Multi-Site Research

N/A: (This is not a multi-site study where the PI is the lead investigator; all sub-sections are omitted as per ETHOS instructions.).

24.0 Coordinating Center Research

N/A: (This study does not involve a coordinating center. The University of Minnesota is the sole site conducting and overseeing all aspects of the research.).

25.0 Resources Available

25.1 Resources Available:

The following resources are available to conduct this research:

1. **Student Investigator Support:** The Student Investigator (James Brandt) is conducting this research as part of their Master's thesis under the direct supervision and mentorship of the Faculty Advisor (Dr. Cecilia Xi Wang). The Faculty Advisor will provide guidance on study design, ethical conduct, data analysis, and interpretation.
2. **Recruitment Feasibility:** Participants will be recruited from the Prolific platform, which has a large pool of US-based adult users. The target sample size of 162 is considered feasible to recruit within an estimated 1-week period, given typical response rates for appropriately compensated online studies.
3. **Time Commitment:** The Student Investigator will dedicate sufficient time for study setup, data collection monitoring, data analysis, and report writing.
4. **Facilities/Technical Resources:**
    a. University of Minnesota's Qualtrics instance for secure survey administration and consent documentation.
    b. The custom-built Kagami.Chat web application (React frontend, Python/FastAPI backend using OpenAI APIs) for chatbot interaction.
    c. Secure University of Minnesota approved cloud storage (e.g., UMN Enterprise Google Drive) for data storage.
    d. Access to statistical software (e.g., SPSS, Python, R) for data analysis.

5. **Financial Resources:** Funding for participant compensation via Prolific is available through personal funds allocated for thesis research.
6. **Availability of Psychological Resources:** While the study is minimal risk, the consent form and pre-survey (if asking about mental well-being) will include contact information for the National Suicide Prevention Lifeline. Participants are also generally expected to have access to their own support systems if needed for issues unrelated to or incidentally arising from this minimal risk research.
7. **Research Team Training:** The Student Investigator is knowledgeable about the protocol and research procedures. The Faculty Advisor provides oversight to ensure all procedures are conducted ethically and appropriately.

26.0 References

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27.0 Health Information and Privacy Compliance

☒I assert that I am not using protected health information (PHI), am not collecting, transmitting, or storing health data internationally, and do not have an appointment in the University of Minnesota Health Care Component. (If you select this box, skip to the next section).

27.1 Health Care Component:

Are any research personnel working on this study part of the Health Care Component (HCC)?

☐ Yes

☒ No

27.2 Select which of the following is applicable to your research:

☒ My research does not require access to individual health information and therefore assert HIPAA does not apply.

☐ I am requesting that all research participants sign a HIPCO approved HIPAA Disclosure Authorization to participate in the research (either the standalone form or the combined consent and HIPAA Authorization).

☐ I am requesting the IRB to approve a Waiver or an alteration of research participant authorization to participate in the research.

Appropriate Use for Research:

☐ An external IRB (e.g. Advarra) is reviewing and we are requesting use of the authorization language embedded in the template consent form in lieu of the U of M stand-alone HIPAA Authorization. Note: External IRB must be serving as the privacy board for this option.

27.3 Preparatory to Research. Please attest to one of the following statements:

☐ I will <u>only</u> be accessing participant medical information/records for the purpose of Preparatory to Research Activities

☐ I will be accessing participant medical information/records beyond the purposes of Preparatory to Research Activities

☐ I am unsure and require HIPCO guidance to determine if the activities I am proposing are considered Preparatory to Research Activities

☒ Not applicable to this study

27.4 Identify the source of Private Health Information you will be using for your research (check all that apply):

☐ I will use the Informatics Consulting Services (ICS) available through CTSI (also referred to as the University's Information Exchange (IE) or data shelter) to pull records for me.

☐ I will collect information directly from research participants.

☐ I will use University services to access and retrieve records from the Bone Marrow Transplant (BMPT) database, also known as the HSCT (Hematopoietic Stem Cell Transplant) database.

☐ I will pull records directly from EPIC.

☐ I will retrieve record directly from axiUm / MiPACS

☐ I will receive data from the Center for Medicare/Medicaid Services

☐ I will receive a limited data set from another institution.

☐ I will receive a de-identified data set from another institution.

☒ Other. Describe: N/A

27.5 Explain how you will ensure that only records of patients who have agreed to have their information used for research will be reviewed:
N/A

27.6 Approximate number of records required for review:
N/A

27.7 Please describe how you will communicate with research participants during the course of this research. Check all applicable boxes:

☐ This research involves record review only. There will be no communication with research participants.

☐ Communication with research participants will take place in the course of treatment, through MyChart, or other similar forms of communication used with patients receiving treatment.

☒ Communication with research participants will take place outside of treatment settings. If this box is selected, please describe the type of communication and how it will be received by participants.

Communication will be entirely online.

Recruitment: Via Prolific study posting.

Consent & Surveys: Via REDCap.

Chat Interaction: Real-time text chat via the Kagami.Chat web application.

Debriefing: Provided on-screen at the end of the study.

Post-Study Contact (Optional, Participant-Initiated): Participants may contact the PI via email (provided in consent/debriefing).

☐ Communication may require the use of interpreter service(s) or translation service(s).

27.8 Explain how the research team has legitimate access to patients/potential participants:

28.0 Health Science Technology (HST) HIPAA Compliant Devices and Data Storage

☒ I assert that I am not using protected health information (PHI) and do not have an appointment in the University of Minnesota Health Care Component.

28.1 HST Device Number
Other non-HST managed devices:

☐ UMP Computer(s)

☐ Store ☐ Analyze ☐ Share

☐ Fairview Computer(s)

☐ Store ☐ Analyze ☐ Share

☐ Other non-HST managed device(s):

28.2 Location(s) of storage, sharing and analysis of research data, including any links to research data (check all that apply):

☐ In the data shelter of the Information Exchange (IE)

☐ Store ☐ Analyze ☐ Share

☐ In the Bone Marrow Transplant (BMT) database, also known as the HSCT (Hematopoietic Stem Cell Transplant) Database

☐ Store ☐ Analyze ☐ Share

☐ In REDCap (recap.ahc.umn.edu)

☐ Store ☐ Analyze ☐ Share

☒ In Qualtrics (qualtrics.umn.edu)

☒ Store ☒ Analyze ☐ Share

☐ In OnCore (oncore.umn.edu)

☐ Store ☐ Analyze ☐ Share

☐ In the University's Box Secure Storage (box.umn.edu)

☐ Store ☐ Analyze ☐ Share

☐ Sponsor Electronic Data Capture Tool (i.e. Advarra or other)

☐ Store ☐ Analyze ☐ Share

☐ In UMP devices/servers

☐ Store ☐ Analyze ☐ Share

☐ In Fairview devices/servers

☐ Store ☐ Analyze ☐ Share

☐ In an AHC-IS supported server. Provide folder path, location of server and IT Support Contact:

☐ Store ☐ Analyze ☐ Share

☒ Other. I will use a server not previously listed. Describe: UMN Enterprise Google Drive (drive.google.com, UMN enterprise account)

☒ Store ☒ Analyze ☐ Share

28.3 Consultants. Vendors. Third Parties:
The following third-party services/technologies will be used in this research:
OpenAI API

1. **Chat Completions Endpoint (/v1/chat/completions):** The gpt-4.1-nano model will be used via this endpoint for generating Kagami's dialogue. User chat inputs will be sent to this endpoint.
2. **Image Edits Endpoint (/v1/images/edits):** The gpt-image-1 model will be used via this endpoint for generating custom avatars. A base image and user-provided text prompts will be sent to this endpoint.
3. **Data Handling by OpenAI:** According to OpenAI's API data usage policy (effective March 1, 2023), data sent to these API endpoints is not used to train or improve OpenAI models (this study will not opt-in to data sharing for training).
4. **Abuse Monitoring & Retention:** OpenAI retains API inputs and outputs from these endpoints for up to 30 days for abuse monitoring purposes. This study will operate under this default data retention policy. For these specific endpoints, OpenAI indicates no persistent application state retention beyond the abuse monitoring logs for the core function of these calls.
5. **Data Transmitted:** For chat, only the current user text turn and de-identified chat history (for context) are sent. For avatar generation, only the user's text prompt and a base image are sent. No other identifiable participant data collected in Qualtrics (e.g., demographics, Prolific ID) will be transmitted directly to OpenAI through these API calls.
6. **Security:** Communication with the OpenAI API will occur over secure HTTPS connections.

Prolific (www.prolific.com)

1. **Purpose:** Prolific will be used solely as a third-party platform for recruiting eligible participants from the United States.
2. **Data Handling:** The research team will receive anonymized Prolific IDs from the platform, which are used for participant payment reconciliation. No other identifiable data about participants is received directly from Prolific. Prolific operates under its own privacy policy and terms of service.

Other Vendors:

No other consultants, vendors, or third-party applications that will have access to or store research data are anticipated for this study, other than standard University of Minnesota IT infrastructure (Qualtrics, UMN Enterprise Google Drive)..

28.4 Data Ownership:

☒ UMN

☐ UMP

☐ Fairview

☐ Sponsor

☐ Third-party university

☐ Other (specify)

28.5 Links to identifiable data:

A unique, random participant code (pid) will be generated in Qualtrics and passed via URL to the Kagami.Chat application to link pre-survey, chat interaction, and post-survey data for each participant. Prolific IDs are collected by Prolific for payment and are provided to the researcher; these will be stored in a separate, secure file temporarily for participant tracking and compensation reconciliation. Prolific IDs will be removed from the primary research dataset and replaced by the pid prior to main data analysis. The separate file linking Prolific ID to pid will be securely destroyed after data collection and compensation are finalized (e.g., within 3–6 months of study completion).

28.6 Sharing of Data with Research Team Members:

Data will be shared between the Student Investigator (James Brandt) and the Faculty Advisor (Dr. Cecilia Wang) via secure UMN systems (e.g., UMN Enterprise Google Drive with restricted access).

28.7 Storage of Documents:

Electronic research documents (protocol, consent forms, survey instruments, de-identified datasets, analysis scripts) will be stored on secure, password-protected UMN Enterprise Google Drive, accessible only to the research team. Any physical documents (not anticipated for this fully online study) would be stored in a locked filing cabinet in a secure office at the University.

28.8 Disposal of Documents:

Electronic data and documents will be securely deleted from storage systems after the retention period (minimum 3 years post-study completion/publication) according to UMN data disposal guidelines. This includes shredding of any physical documents (not anticipated).